Prabhavit Gupta

Agentic AI Engineer | Autonomous Agent Architectures, Model Context Protocol (MCP) & Evals

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Executive Summary

- **Production Agent Architecture & Tool Orchestration:** Software engineer with 3 years commercial experience architecting and shipping autonomous AI agents, tool loops, and production LLM systems. Hands-on experience developing multi-turn agent workflows with LangGraph and native Python state machines, balancing quality, latency, and token economics.
- **Model Context Protocol (MCP) & Context Management:** Deep practical experience designing custom Model Context Protocol (MCP Server & Client) tools and interfaces, standardizing secure agent communication with relational databases and external analytical APIs. Proficient in dynamic context pruning, memory persistence, and structured output enforcement.
- **Eval-Driven Engineering & System Observability:** Strong track record establishing rigorous evaluation frameworks (RAGAS, automated regression test suites) to quantify agent performance and eliminate non-deterministic regressions. Experienced instrumenting distributed tracing (Langfuse, OpenTelemetry), containerized microservices (Docker), and automated CI/CD pipelines.

Technical Skills

- **Agentic AI Systems:** Autonomous Agents, Tool Loops, Function Calling, ReAct Reasoning, Multi-Agent Collaboration, LangGraph, LangChain, Stateful Execution, Human-in-the-Loop Approval Gates.
- **Model Context Protocol (MCP) & LLMs:** Model Context Protocol (MCP Server & Client SDKs), Claude 3.5 Sonnet, GPT-4o, Prompt Engineering, Structured Outputs (Pydantic / JSON Schema), Model Routing.
- **Evaluation & Observability:** Eval-Driven Development, RAGAS Benchmarking, Regression Detection Test Suites, Langfuse Tracing, OpenTelemetry, Token Budgeting, Latency Profiling.
- **Backend & Data Infrastructure:** Python (AsyncIO, FastAPI, PyTest, Typing), TypeScript, PostgreSQL (pgvector), Redis (Caching & Queues), Docker, Microservices, REST / WebSocket APIs.
- **Software Quality & Practices:** Automated Testing, Defensive Architecture, Code Reviews, ADRs, Secure Credential Isolation, Git, GitHub Actions CI/CD.

Professional Experience

AI Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Architected and shipped production agentic systems and tool-using LLM platforms for European client applications, taking features from architecture design through live deployment.
- Developed custom Model Context Protocol (MCP) servers and clients, exposing database query tools, analytical functions, and third-party APIs to autonomous agents via standardized protocols.
- Built resilient multi-step agent orchestration loops with state persistence, context window management, automated retries, and fallback mechanisms for non-deterministic model failures.
- Implemented comprehensive offline and online evaluation pipelines using RAGAS, measuring tool-calling precision, reasoning faithfulness, and task success rates before deploying prompt updates.
- Deployed distributed tracing using Langfuse and OpenTelemetry, establishing real-time visibility into token usage, latency distributions, and agent decision paths.
- Containerized AI microservices using Docker and automated regression testing pipelines via GitHub Actions, maintaining high release velocity and reliability.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Built asynchronous backend services in Python and TypeScript for high-volume automated data extraction, parsing, and classification.
- Implemented semantic search and vector retrieval pipelines with PostgreSQL and Redis, ensuring reliable sub-second data access for downstream processing.
- Created comprehensive test suites with PyTest and CI/CD automation, ensuring high test coverage and preventing regressions across distributed services.
- Collaborated with cross-functional product and technical stakeholders to translate quantitative domain requirements into robust, modular software architectures.

Key Engineering Projects

- **Autonomous Tool-Using Agent Platform (Python + LangGraph + MCP + Postgres):** Multi-agent collaborative execution environment running verified tool operations via Model Context Protocol against enterprise databases, featuring state preservation and Langfuse tracing.
- **High-Throughput Financial Document & Data Engine (FastAPI + pgvector):** Retrieval and structured data extraction microservice with automated validation schemas, semantic caching, and regression evaluation harnesses.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous foundation in formal logic, statistical reasoning, decision theory, and complex systems analysis.
- **Languages:** English (C1 Professional, fluent written/spoken), Italian (Working proficiency).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to live and work across the European Union without sponsorship; available immediately for remote engagement.